An Introduction to Discrete Probability

Section 7.1

Probability

- Probability is basically about answering:
  - "How likely is something to happen?"
- For example:
  - What is the chance of getting heads when flipping a coin?
  - What is the chance of rolling a 6 on a die?
  - What is the chance it rains tomorrow?

Sample Space

- The **sample space** is the list of *all possible outcomes*.
- Usually written as ***S***.
  - Example 1: coin flip
    - $S = \{\text{Heads}, \text{Tail}\}$
  - Example 2: rolling a die
    - $S = \{1, 2, 3, 4, 5, 6\}$
- The sample space contains everything that could happen.

Event

- An **event** is just a group of outcomes we care about.
- **Example:** Roll a die and ask:
  - "What is the chance of rolling an even number?"
- Even numbers are:

$$E = \{2, 4, 6\}$$

- This set is the event.

- **Another example:**
  - "What is the chance of rolling a number greater than 4?"

$$E = \{5, 6\}$$

- Probability measures how likely the event is.

Laplace's idea was:

- If all outcomes are likely to occur, then

$$p(E) = \frac{|E|}{|S|}$$

- This means:

$$\text{Probability} = \frac{\textit{Number of favorable outcomes}}{\textit{Total outcomes}}$$

Example

Flip a coin.

- Sample space:

$$S = \{\text{Heads}, \text{Tail}\}$$

- Event: Getting heads

$$E = \{Heads\}$$

Now count:

  - Favorable outcomes = 1
  - Total outcomes = 2

- So:

$$p(E) = \frac{1}{2}$$

Why Probability Is Always Between 0 and 1

- Probability can never be:
  - less than 0
  - greater than 1
- Because:
  - 0 means impossible
  - 1 means guaranteed

Examples:

- Probability the sun rises tomorrow ≈ very close to 1
- Probability of rolling a 7 on a normal die = 0
- So:

$$0 \le p(E) \le 1$$

Applying Laplace's Definition

Example: An container contains four blue balls and five red balls. What is the probability that a ball chosen from the container is blue?

Step 1: Count total balls

$$4 + 5 = 9$$

- So there are 9 possible balls you could pick.

Step 2: Count favorable outcomes

- We want a **blue** ball.
- There are 4 blue balls.
- So:
  - Favorable outcomes = 4
  - Total outcomes = 9

Step 3: Use probability formula

$$p(E) = \frac{|E|}{|S|}$$

$$p(blue) = \frac{4}{9}$$

Applying Laplace’s Definition

Example: What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

Step 1: Understand the outcomes

- Each die has 6 possible numbers:

  1,2,3,4,5,6

- When rolling **two dice**, we combine the results.
- Examples:
  - **Find outcomes that give sum = 7**
  - These are:
    - (1, 6)
    - (2, 5)
    - (3, 4)
    - (4, 3)
    - (5, 2)
    - (6, 1)
  - There are 6 favorable outcomes.

Step 2: Count total outcomes

- First die: 6 possibilities
- Second die: 6 possibilities

- Multiply them:

$$6 \times 6 = 36$$

- So there are 36 total possible outcomes.

Step 3: Use probability formula

$$p(sum = 7) = \frac{6}{36}$$

- Simplify:

$$\frac{6}{36} = \frac{1}{6}$$

Example 3:
In a lottery, a machine randomly picks a 4-digit number.
A player also chooses a 4-digit number.

The player wins the prize if:

- all four digits match exactly
- and the digits are in the correct order
- What is the probability that the player wins the prize?

Lottery Example Simplified

A player must guess a **4-digit number** correctly.

Example winning number: 5271

- To win:
  - all 4 digits must match
  - and they must be in the correct order
- So:
  - 5271 = win
  - 5712 = lose (wrong order)

Step 1: Count all possible 4-digit numbers

Each digit can be:

$$0,1,2,3,4,5,6,7,8,9$$

That means:

- 10 choices for the first digit
- 10 choices for the second digit
- 10 choices for the third digit
- 10 choices for the fourth digit

Multiply them:

$$10 \times 10 \times 10 \times 10$$
$$= 10^4$$
$$= 10000$$

Step 2: Count winning outcomes

Only **one** number matches exactly.

So:

favorable outcomes = 1

total outcomes = 10000

Step 3: Use probability formula

$$p(E) = \frac{1}{10000}$$

So the probability of winning is:

$$\boxed{\frac{1}{10000}}$$

- **Question:**
  In a lottery, a machine randomly picks a 4-digit number. A player also chooses a 4-digit number.
- The player wins a **small prize** if exactly **three digits match** the winning number in the correct positions.
- What is the probability that the player wins the small prize?

Step 1: Total possible outcomes

A 4-digit number has:

- 10 choices for each digit (0–9)

So total numbers:

$$10 \times 10 \times 10 \times 10 = 10^4 = 10000$$

Step 2: What does "exactly 3 digits match" mean?

It means:

- Out of 4 positions, **3 digits are correct**
- 1 digit is wrong

So we choose:

- which 3 positions are correct
- and the 1 wrong position

Step 3: Choose the wrong position

There are 4 positions, so:

$$4 \ choices$$

Step 4: Wrong digit choices

For the wrong position:

- any digit except the correct one
- so 9 choices

Step 5: Total favorable outcomes

$$4 \times 9 = 36$$

So there are **36 ways** to match exactly 3 digits.

Step 6: Probability

$$p = \frac{favorable}{total} = \frac{36}{10000} = \frac{9}{2500}$$

Applying Laplace's Definition

- **Example**: In a lottery, a player chooses **6 numbers** from the numbers **1 to 40**.
- A machine also randomly selects **6 winning numbers** from 1 to 40.
- The order does not matter.
- What is the probability that the player picks all 6 correct numbers?

Step 1: Total possible ways to choose 6 numbers from 40

We are just selecting 6 numbers, not caring about order.

So this is a combination problem:

$$\text{Total } ways = \binom{40}{6}$$

$$\binom{40}{6} = 3{,}838{,}380$$

Step 2: Favorable outcomes

Only **1** set of numbers is the winning set.

So:

favorable outcomes = 1

Step 3: Probability

$$p = \frac{1}{3{,}838{,}380}$$

There is **only one correct combination** out of **3.8 million possible combinations**, so the chance of winning is extremely small.

The Probability of Complements and Unions of Events

Theorem 1: Let E be an event in sample space S. The probability of the event $\overline{E} = S - E$, the complementary event of E, is given by

$$p(\overline{E}) = 1 - p(E).$$

Proof: Using the fact that

$$p(\overline{E}) = \frac{|S| - |E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E).$$

The Probability of Complements and Unions of Events

- E = **what you want**
- $\overline{E}$ = **what you don't want**
- Together they make the whole sample space.

$$p(\overline{E}) = 1 - p(E).$$

- Suppose:
- Probability of rain today = 0.3
- Then:
- Probability of NOT rain =

$$1 - 0.3 = 0.7$$

The Probability of Complements and Unions of Events

- **Example**: A fair coin is flipped once.
  What is the probability that the result is **not heads**?

The Probability of Complements and Unions of Events

- **Example**: A fair coin is flipped once.
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Step 1: Possible outcomes

$$S = \{\text{Heads}, \text{Tail}\}$$

Step 2: Event we want

We want:
"not heads"
That means:

$$\{Tails\}$$

Step 3: Count probabilities

There are 2 equally likely outcomes:
Heads
Tails
So:
Probability of Heads = 1/2

Step 4: Use complement idea

Instead of directly thinking "not heads", we do:

$$P(not\ Heads) = 1 - P(Heads)$$
$$= 1 - \frac{1}{2}$$
$$= \frac{1}{2}$$

The Probability of Complements and Unions of Events

Theorem 2: Let E_1 and E_2 be events in the sample space S. Then

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

Why do we subtract something?

Because:

- If both events happen together, we **count them twice**
- So we subtract the overlap once

Example

Suppose:

- 10 students like Math
- 8 students like Physics
- 3 students like both

- Total students = 20

The Probability of Complements and Unions of Events

Example

Suppose:

- 10 students like Math
- 8 students like Physics
- 3 students like both
- Total students = 20

Step 1: Add both groups

$$10 + 8 = 18$$

But this includes the 3 students twice.

Step 2: Fix double counting

$$18 - 3 = 15$$

So:

•15 students like Math OR Physics

Step 3: Probability

$$P = \frac{15}{20} = \frac{3}{4}$$

Note: “Add both probabilities, but subtract the overlap so you don’t count it twice”

The Probability of Complements and Unions of Events

Example: A number is randomly chosen from **1 to 100**. What is the probability that the number is: divisible by **2 OR 5?**

The Probability of Complements and Unions of Events

Example: A number is randomly chosen from **1 to 100**.

What is the probability that the number is: divisible by **2 OR 5?**

Step 1: Total outcomes

All numbers from 1 to 100:

$$S = 100$$

Step 2: Count multiples of 2

Numbers divisible by 2:

$$2,4,6,\ldots,100$$

$$\frac{100}{2} = 50$$

So:

•divisible by 2 = 50 numbers

Step 3: Count multiples of 5

Numbers divisible by 5:

$$5,10,15,\ldots,100$$

$$\frac{100}{5} = 20$$

So:

•divisible by 5 = 20 numbers

Step 4: Fix double counting

Some numbers are counted twice (divisible by both 2 and 5).

That means divisible by 10:

$$10,20,30,\ldots,100$$

$$\frac{100}{10} = 10$$

So:

•overlap = 10 numbers

Step 5: Use formula

$$(2\ or\ 5) = 50 + 20 - 10$$
$$= 60$$

Step 6: Probability

$$P = \frac{60}{100}$$

$$= \frac{3}{5}$$